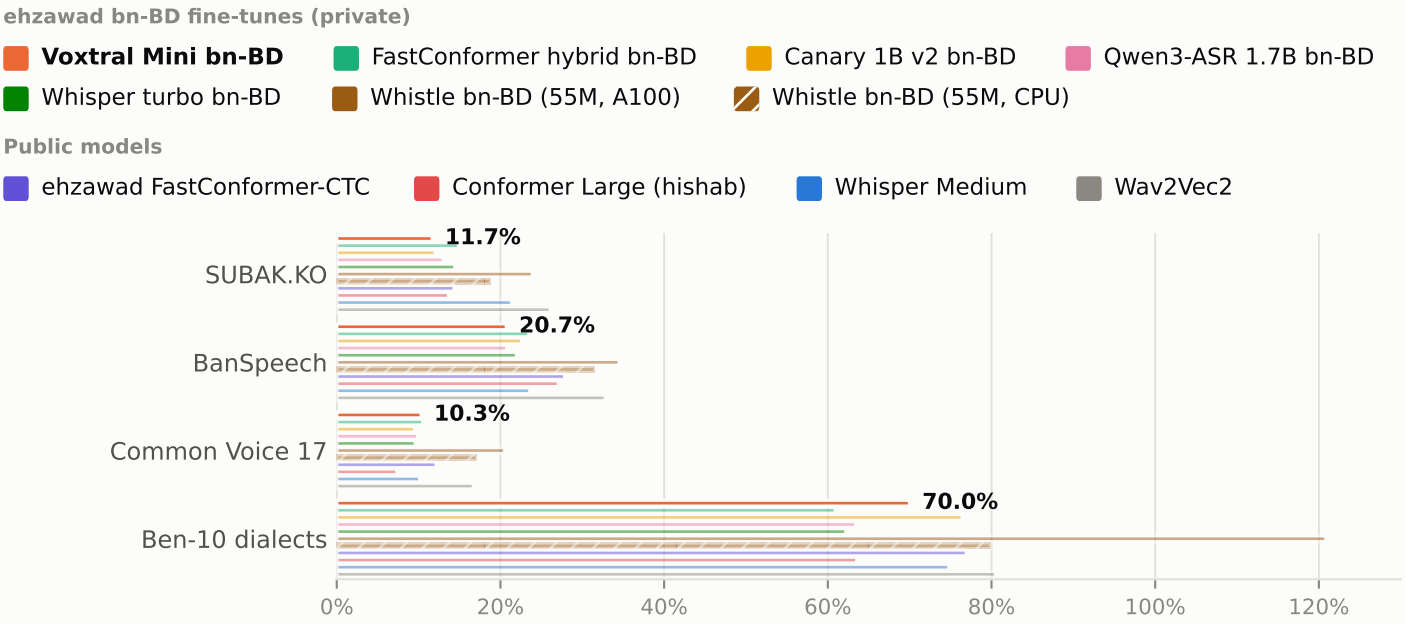
Voxtral Mini bn-BD

Bangladeshi and crowd-sourced test sets · word error rate, lower is better · Voxtral Mini in orange



The numbers

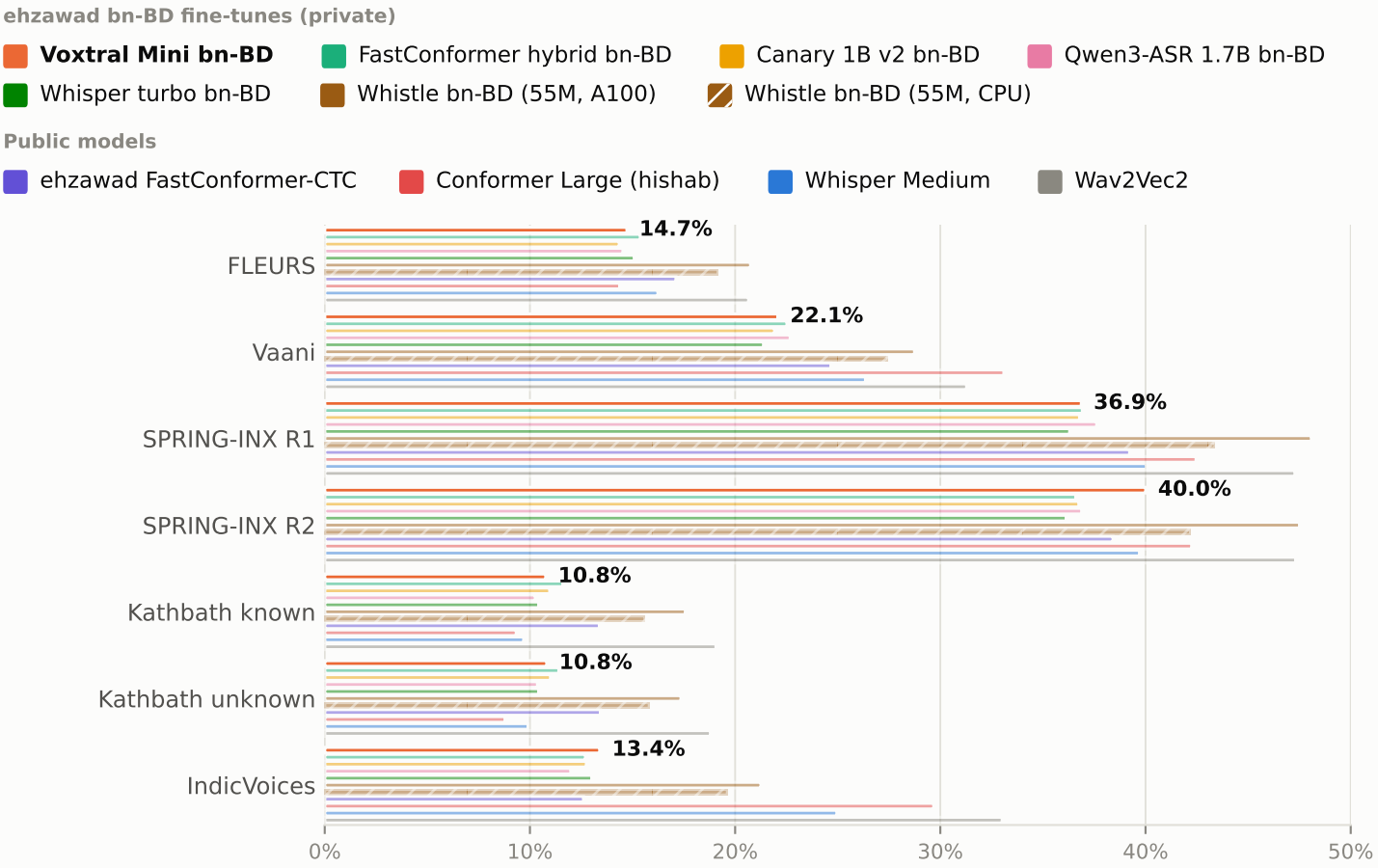
	SUBAK.KO	Ban-Speech	Common Voice 17	Ben-10 dialects
Voxtral Mini bn-BD	**11.7**	**20.7**	**10.3**	**70.0**
FastConformer hybrid bn-BD	14.9	23.5	10.5	**60.9**
Canary 1B v2 bn-BD	12.0	22.6	9.5	76.4
Qwen3-ASR 1.7B bn-BD	13.0	20.8	9.9	63.4
Whisper turbo bn-BD	14.4	22.0	9.6	62.2
Whistle bn-BD (55M, A100)	23.9	34.5	20.5	120.9
Whistle bn-BD (55M, CPU)	18.8	31.5	17.1	79.9
ehzawad FastConformer-CTC	14.3	27.8	12.1	76.9
Conformer Large (hishab)	13.6	27.1	**7.3**	63.5
Whisper Medium	21.4	23.6	10.1	74.8
Wav2Vec2	26.1	32.8	16.7	80.5

lowest word error rate on that set

- Voxtral Mini has the lowest word error rate on SUBAK.KO and BanSpeech.
- 288 of SUBAK.KO's 1,000 references write numbers as digits, which models that write numbers as words cannot match; on the other recordings Voxtral Mini reads 5.4%.
- SUBAK.KO and Common Voice are not clean head-to-heads: ehzawad FastConformer-CTC and Conformer Large trained on both corpora, and FastConformer-CTC's Common Voice data may include these very clips.
- Ben-10 is regional dialect speech written in dialect spelling; its test split is private, so this is its validation split. Common Voice records no speaker region. The private models' training data is undocumented.

Indian Bengali test sets

Word error rate, lower is better · Voxtral Mini in orange



The numbers

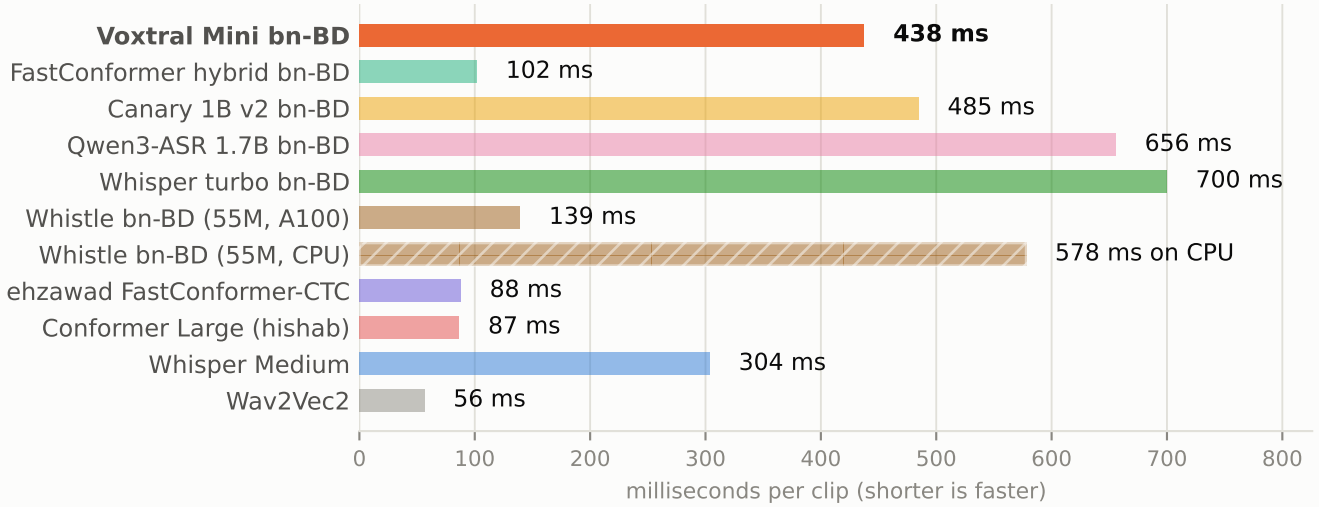
	FLEURS	Vaani	SPRING R1	SPRING R2	Kathbath known	Kathbath unknown	Indic Voices
Voxtral Mini bn-BD	**14.7**	**22.1**	**36.9**	**40.0**	**10.8**	**10.8**	**13.4**
FastConformer hybrid bn-BD	15.4	22.5	36.9	36.6	11.6	11.4	12.7
Canary 1B v2 bn-BD	**14.4**	21.9	36.8	36.8	11.0	11.0	12.8
Qwen3-ASR 1.7B bn-BD	14.5	22.7	37.6	36.9	10.3	10.4	**12.0**
Whisper turbo bn-BD	15.1	**21.4**	**36.3**	**36.1**	10.4	10.4	13.0
Whistle bn-BD (55M, A100)	20.8	28.7	48.1	47.5	17.6	17.4	21.3
Whistle bn-BD (55M, CPU)	19.2	27.4	43.4	42.2	15.6	15.8	19.6
ehzawad FastConformer-CTC	17.1	24.7	39.2	38.4	13.4	13.4	12.6
Conformer Large (hishab)	14.4	33.1	42.5	42.2	**9.3**	**8.8**	29.7
Whisper Medium	16.2	26.4	40.0	39.7	9.7	9.9	25.0
Wav2Vec2	20.7	31.3	47.3	47.3	19.1	18.8	33.0

lowest word error rate on that set

- Kathbath and IndicVoices are not clean head-to-heads: ehzawad FastConformer-CTC trained on other splits of both, and Conformer Large on Kathbath.
- The private models' training data is undocumented, so overlap with any of these sets cannot be ruled out.

Speed, method and caveats

Time per FLEURS test clip (13.4 s of audio), one at a time, on one A100 (Whistle bn-BD also on CPU)



Voxtral Mini takes 0.44 s for a 13-second clip, about 31× faster than real time. Faster: FastConformer hybrid bn-BD, Whistle bn-BD (55M, A100), ehzawad FastConformer-CTC, Conformer Large (hishab), Whisper Medium and Wav2Vec2. Slower: Canary 1B v2 bn-BD, Qwen3-ASR 1.7B bn-BD and Whisper turbo bn-BD. The times cover each model's whole pipeline, audio loading included: vLLM for Voxtral Mini bn-BD, Qwen3-ASR 1.7B bn-BD, Whisper turbo bn-BD and Whisper Medium, NeMo or PyTorch for the rest, so they compare setups, not just the models. Whistle bn-BD appears twice: on an A100 in JAX (greedy), comparable with the other bars, and hatched on one CPU core in its needle engine, the on-device cost.

How it was measured

- One Colab A100 (40 GB) on 13 Oct 2026, one model on the GPU at a time; Whistle bn-BD on an A100 (80 GB) and a CPU on 21-22 Oct.
- The same audio for every model: 16 kHz, cut at pauses into pieces of at most 25 s.
- Plain decoding: greedy, one clip at a time, no external language model (Whistle bn-BD on CPU: its engine's default decoding).
- FLEURS: the 920 test recordings. Other sets: 1,000 recordings each, a seeded sample of each set's official split.
- One scorer for everyone: punctuation removed and Bengali text Unicode-normalised before counting errors.

Keep in mind

- Seven of the eleven sets are Indian Bengali; SUBAK.KO, BanSpeech and Ben-10 are Bangladeshi.
- About half of SPRING-INX's reference sentences contain English words written in Latin letters, which no model reproduces, so SPRING error rates are inflated for every model alike. Whistle bn-BD leaks English fragments into 18/15 (A100) and 26/18 (CPU) of 1,000 R1/R2 transcripts.
- On SPRING-INX R2, Voxtral Mini got stuck repeating one word on 3 recordings; those alone account for 663 of its 8,143 word errors there, and they stay in its score.
- Like Conformer Large and Whisper Medium, Voxtral Mini writes numbers as Bengali words, which counts as an error wherever the reference uses digits.
- Whisper's tokenizer spends about 13 tokens per Bengali word, so Whisper turbo's longest pieces were cut at its 448-token limit: 403 of Ben-10's 1,000 recordings.